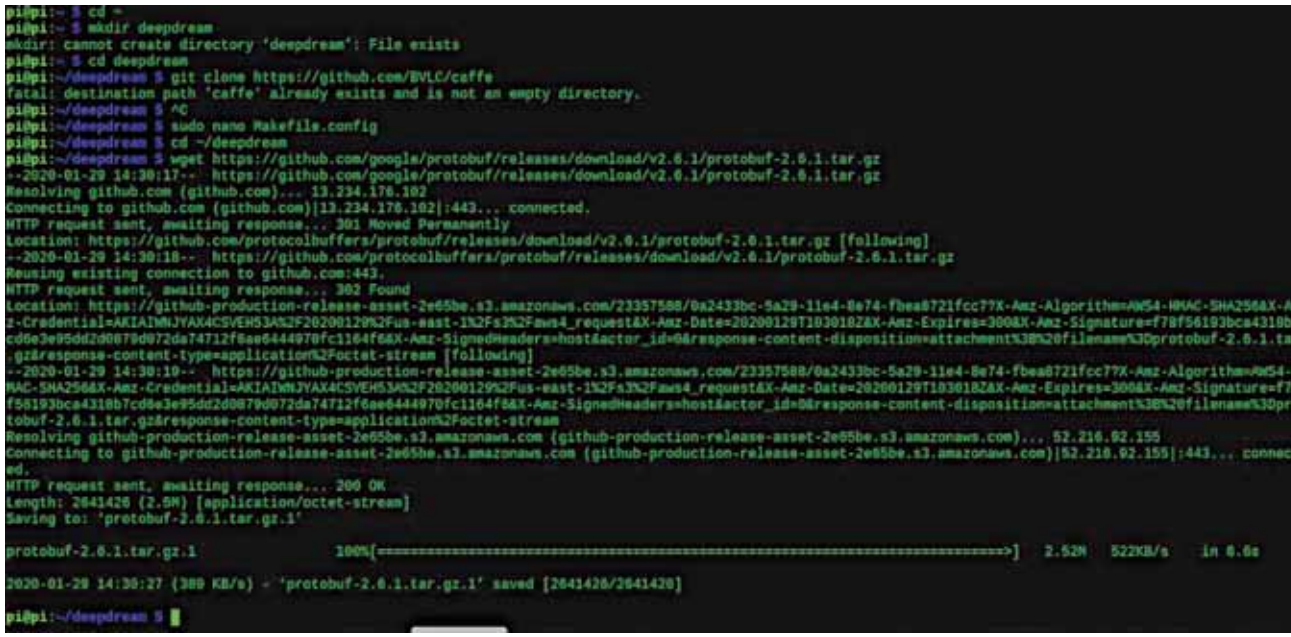

Figure 6: Cloning Deedream

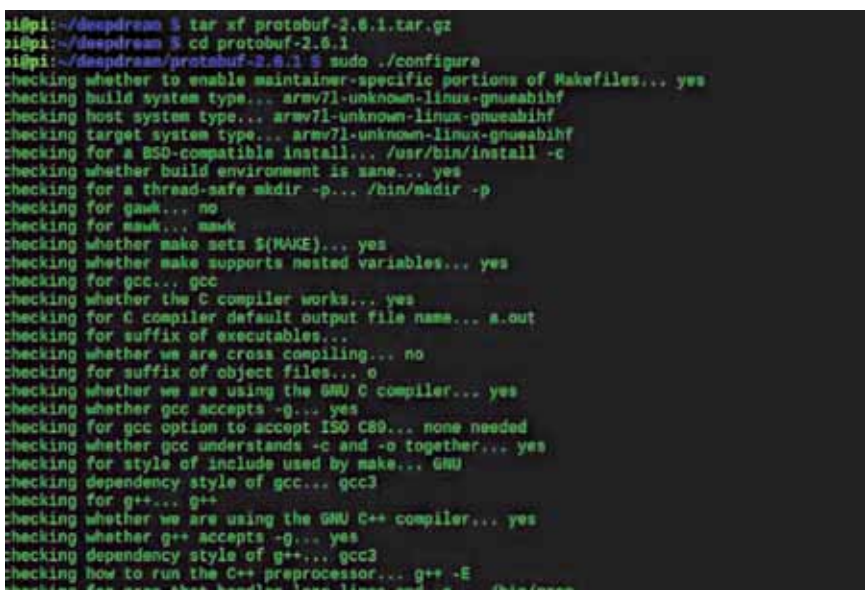

Figure 7: Installing protobuf

```
libavformat-dev libswscale-dev libv4l-
dev
sudo apt-get install libxvidcore-
  dev libx264-dev
sudo apt-get install libgtk2.0-dev
  libgtk-3-dev
sudo apt-get install libatlas-base-
  dev gfortran
```

After that swap the file size. To do so, run the following command:

```
sudo nano /etc/dphys-swapfile
```

Then change the line CONF_SWAPSIZE=100 to CONF_SWAPSIZE=1024

```
sudo /etc/init.d/dphys-swapfile stop
sudo /etc/init.d/dphys-swapfile start
sudo pip3 install opencvsudo pip3
install
   numpy
wget https://bootstrap.pypa.io/get-pip.
py
pip install dlib
```

Figure 4 shows how to install dlib required for face recognition.

Next, to install the face recognition library, run the following command in the terminal:

```
sudo pip3 install face recognition
```

Run the *sudo apt-get update* command again; with that the basic setup for emotion recognition is done.

For deep dream requirements/setup, install the following modules in Python:

- Caffe
- Protobuf (also known as Protocol Buffers)
- PsyCam

Open your Linux terminal and run the above commands to install the modules (refer Figure 5 through Figure 11). You can also follow the Google Deep Dream official site for installation on Raspberry Pi.

Coding

Having completed the basic setup of the libraries, write the code for real-time face recognition. First, download OpenCV, OpenCV